# Carmen

## Research, pedagogy and product strategy

**Version 2.0 | 9 November 2026 | Foundation-Year 6 with variable complexity and older pathways**

### The research changes the innovation claim

AI bandmates, tempo-following accompaniment, gesture-controlled instruments, adaptive questions and generated teaching examples already have documented precedents. Moises/MusicFirst and Soundverse are particularly relevant overlaps [U01-U02; U07].

This review does not establish an exclusive USP. Carmen needs a demonstrated advantage for a defined learner, teacher or purchaser. A combination of useful features can support a product, but it is not evidence that nobody else offers it.

### The candidate product to test

Carmen is proposed as an accessible music-learning environment built around learner-owned music. A real local model proposes a bounded musical response to the learner's phrase and a teaching objective. A separate music compiler verifies the declared symbolic change before playback through authentic samples. The learner predicts, listens, directs, rejects or revises, then tries a fresh independent task. Teacher evidence distinguishes original work, AI contribution, access support and independent performance.

The scope is the full primary suite from Foundation-Year 6, with meaningful development across four bands. Secondary and adult learners can use suitable contexts and greater conceptual depth. The ten foundation lessons and new band missions are drafts, not a completed curriculum catalogue.

### Build something concrete

The MVP should demonstrate two sampled instruments, four distinct primary starter pathways and an older extension. Start with short musical replies between phrases and a compact local response-planning model. Optional camera gestures are an experiment with touch/keyboard equivalents, disabled by default. Do not add cloud child-data processing or a headset requirement.

### What the pack now provides

The updated Claude Code pack contains an explicit competitor matrix, a novelty verdict, model/device gates, a musical reply contract, illustrative validation fixtures, primary progression and a claim-status register. It includes no app execution, trained weights, real-device benchmark, child-pilot results or educator approval.

The technical application is a proposal. Effectiveness, usability, comparative value and commercial sustainability remain questions for the implemented product to answer.

# 1. What the learning evidence supports

## Findings translated into product decisions

| Evidence | Finding within its actual scope | Implication for Carmen |
|---|---|---|
| McPherson's three-year study, 157 beginning instrumentalists [R01] | Practice time and rehearsed notation alone did not adequately describe early development; musical strategies mattered. | Teach ways to listen, isolate a difficulty and test a strategy. Track more than minutes or correct notes. |
| Digital listening study, four children aged 8-10 over 22 weeks [R03] | Customisable listening supported aural problem-solving; sustained teacher facilitation still mattered. | Make replay and comparison central. The small qualitative study does not prove independent app efficacy. |
| Primary-teacher autonomy intervention, 18 teachers and 105 lessons [R04] | Coaching improved verbal autonomy support; musical and bodily changes were uneven. | Offer meaningful choices and responsive demonstrations. A choice menu alone does not establish agency. |
| Randomised violin feedback study, 50 analysed novices [R05] | Feedback improved some movement measures while sound suffered during practice; later sound outcomes were more favourable. | Evaluate the musical outcome, attention demands and later independent performance, rather than maximising corrections. |
| 2026 review of AI and agency in music, 21 studies [R06] | Reports both support and dependence risks. Evidence is concentrated in China and higher education, with limited long-term causal evidence. | Treat younger-child benefits and durable independence as hypotheses requiring direct testing. |
| High-school mathematics field experiment [R07] | AI could improve assisted work while unassisted learning suffered; teacher-designed hint safeguards mitigated the harm. | Measure performance after teaching hints are removed. This is a design warning from another subject. |
| AERO teaching guidance [R09-R11] | Demonstration, scaffolding and varied, spaced practice are useful teaching principles. | Combine short explicit teaching with guided application and later independent checks. |

## Research confidence

The best-supported foundation is the learning process, not an AI brand or a single music method. The literature on digital support for self-regulated instrumental practice remains uneven; Wan and colleagues explicitly identify the need for stronger research [R02].

Carmen therefore needs a distinction between **evidence-informed design** and **evidence of effectiveness**. The former can be built now; the latter must come from evaluating the implemented product.

Music has musical, expressive and cultural value. Broad claims that Carmen will improve IQ, mathematics or general academic attainment are not justified by this evidence. Sala and Gobet's meta-analysis cautions against such far-transfer claims [R08].

# 2. The pedagogical model

## Use musical experience, guided teaching and learner decisions together

| Foundation | Useful contribution | Product expression |
|---|---|---|
| Kodaly-informed practice [P01] | Singing, carefully sequenced learning and encountering sound before symbols. | Let learners hear and reproduce a pattern, then connect it to a representation. Offer alternatives to vocal participation. |
| Orff-informed practice [P02] | Imitation, exploration, improvisation and composition. | A learnt pattern becomes material the student can reshape and combine with others. |
| Gordon's audiation framework [P03] | Developing musical understanding beyond reproducing a visible cue. | Ask learners to recall, predict or compare musical ideas. A correct tap alone cannot establish inner hearing. |
| Self-regulated learning [R01-R03] | Goals, monitoring, strategy selection and reflection. | The learner chooses a focus, attempts it, listens back, selects support and checks again. |
| Explicit instruction and scaffolding [R09-R10] | Clear models and manageable supported practice. | Explain one new concept, demonstrate it musically, practise together and gradually remove teaching hints. |
| Embodied and social music learning [P04] | Gesture, movement, demonstration and interaction have teaching roles. | Include optional movement, body percussion, partner responses and shared performances. |

These are compatible influences, not a claim that all methods are interchangeable or that one is universally superior. Carmen needs a coherent, educator-reviewed sequence rather than a collection of teaching-method labels.

## The activity cycle

1. **Experience:** hear contrasting examples or explore a sound.
2. **Notice:** identify something worth investigating.
3. **Learn:** receive a concise model or explanation when needed.
4. **Try:** play, build, sing or select a musical response.
5. **Reflect:** listen back and compare the result with an intention.
6. **Change:** choose one strategy and retry.
7. **Create and share:** apply the idea in original work and discuss choices.

For an early learner, reflection may be pointing to a preferred version. An older learner may compare interpretations and justify decisions using musical vocabulary. The intellectual action changes in depth; the system need not make every response text-based.

## A spiral across the suite

Revisit rhythm, melody, expression, structure and context with increasing demands. A learner first experiences contrast, then identifies it, controls it, combines it and explains its effect. Later work should use familiar concepts in unfamiliar music and independently chosen projects.

The proposed sequence is a design synthesis. Curriculum alignment and these research influences do not automatically validate every activity.

# 3. The proposed technical contribution

## A musical response that can be inspected

The candidate innovation is a connection between local learned response planning, an explicit music representation, task-specific verification and evidence of independent application. It must not be described as the invention of adaptive music or AI jamming.

| Component | Proposed behaviour | Why it matters |
|---|---|---|
| Learner phrase and intention | Preserve original musical events and the learner's declared goal. | A reply is grounded in actual work. |
| Learned response planning | Select a bounded musical response using the phrase, objective and supported evidence. | AI influences the interaction, rather than adding generic commentary. |
| Musical compilation and checking | Admit only permitted operations and verify declared symbolic invariants. | A fluent explanation cannot disguise the wrong musical example. |
| Learner direction | Let the student predict, reject, redirect and revise. | The learner retains musical authority. |
| Independent evidence | Use a fresh related task with teaching hints removed and access retained. | Assistance is distinguished from understanding. |

## A concrete example

A learner creates four sounds. The task investigates a silent beat. The partner proposes replacing one identified sound with a rest while retaining its time slot. The checker verifies the operation and the unaffected events before playback. The learner compares, chooses a response and applies the idea to new music.

For older or more experienced learners, the same architecture can support motifs, structure or arrangement. The representation and musical demands change separately from motor demands.

## Precision about claims

A checked symbolic difference does not mean only one perceptual property changed. Dynamics can affect acoustic timbre; adding a part can affect harmony and texture. Record precisely what the system verified, and let the learner/teacher judge the musical effect.

A small model trained on synthetic examples can demonstrate computation. It cannot validate an explanation of a child's difficulty or establish a learning advantage.

## What could become defensible

The potential asset is a tested system of reviewed tasks, reliable response decisions and usable school workflows. The pack treats this as a hypothesis. Comparative evaluation and repeat adoption are required before claiming an edge.

# 4. Whole-primary suite

## Eight proposed domains, developed across every primary band

These domains organise Carmen's content; they are not ACARA's strand names. Every band should include listening, making, performing and discussion of musical purpose.

| Domain | Foundation | Years 1-2 | Years 3-4 | Years 5-6 |
|---|---|---|---|---|
| Pulse, rhythm and metre | Sound, silence and a shared beat. | Beat versus rhythm; simple rests and patterns. | Repeated patterns, coordination and contrasting rhythms. | Rhythmic layers, metre and expressive timing. |
| Pitch and melody | High/low and melodic exploration. | Short pitch patterns and musical questions. | Shape, repetition and simple melodic development. | Develop motifs; choose pitches for expressive effect. |
| Dynamics and expression | Explore loud/soft contrasts safely. | Change speed or loudness for a purpose. | Compare interpretations and explain a choice. | Shape phrases and justify expressive decisions. |
| Timbre and texture | Explore and name sound differences. | Select sounds for a musical idea. | Combine parts and identify their roles. | Control balance, layering and instrumental combinations. |
| Form and composition | Create a short sound idea. | Repeat, contrast and combine ideas. | Organise a piece into identifiable sections. | Develop structure using purposeful compositional devices. |
| Listening and contexts | Music at home, school and community. | Explore people, occasions and musical experiences. | Discuss where, how and why music is made. | Examine how music communicates and sustains culture. |
| Performing with others | Share and respond in a supported setting. | Play or sing short pieces together. | Rehearse and present learnt or original music. | Refine a performance for a chosen audience. |
| Representation and reflection | Recall or show a sound idea. | Link sounds with useful visual patterns. | Use suitable notation or recordings; describe changes. | Document music and explain revisions and intentions. |

This proposed progression draws on the band-level references [C03-C07, C12]. It is a content framework, not a claim that the suite is already authored or fully aligned.

## A full suite needs breadth and sequence

Each band needs more than a rhythm path with different artwork. Build projects that connect domains: a soundscape, a rhythmic conversation, a melodic story, an arrangement and a shared performance. Include voice and body-based activities as available options; a screen should not become the sole definition of musicianship.

First Nations content needs authorised material, context and appropriate collaboration. Cultural material must not be scraped into a model or presented as generic instrument decoration [G07].

## Expanding to secondary and adults

Keep older beginners' presentation dignified while allowing advanced reasoning, genre investigation and longer creative projects. Experienced learners can explore harmony, multi-part composition, complex metre and critical interpretation. Secondary and senior assessment coverage require their own content and mapping [C06-C07, C11].

# 5. Variable complexity without age ceilings

## A learner has several forms of competence

| Dimension | Lower demand or more support | Higher demand or greater independence |
|---|---|---|
| Musical material | One sound or short recurring pattern. | Multiple voices, longer structures and interacting ideas. |
| Time | Self-paced construction or slow practice. | Live performance, changing tempo or complex rhythmic relationships. |
| Input | Large pads, single actions, keyboard access. | More simultaneous controls or an optional physical MIDI instrument. |
| Representation | Listen, imitate, icons or a visual sequence. | Notation, score comparison, analysis and documentation. |
| Practice | Model, cue and guided strategy selection. | Learner-designed goals, rehearsal and self-evaluation. |
| Creative task | Change one aspect within a clear frame. | Develop a personally chosen composition and justify its form. |
| Language and context | Short spoken/visual prompts. | Precise vocabulary and analytical discussion appropriate to the learner. |

Input demand is not an achievement ladder. A student using a switch or a large pad may make advanced musical decisions. Similarly, a fluent reader may still be a beginning musician.

## Preserve the goal while changing access

If the goal is to recognise a rhythmic contrast, a timed keyboard response adds an avoidable confound. A listening choice can provide more relevant evidence. If the goal is live rhythmic performance, a sequencer cannot demonstrate that same skill; report the distinction.

Teaching hints can fade as independence grows. Captions, accessible controls, screen-reader support and other access adjustments should remain available. Removing assistance must not mean removing access.

## Keep expression under learner control

Offer choices over sound, musical material, artistic intention and strategies. A learner may decline a recommendation or ask for another example. For a fixed copying exercise, describe a departure from the target accurately; for open creation, use the learner's intention and the task's stated constraints.

Avoid assigning a universal creativity score, inferring learning styles, diagnosing disability or treating equal-tempered piano conventions as the definition of every musical tradition.

## How to build it credibly

Use educator-authored task families with explicit parameters and tested constraints. Adapt from an evidence profile that distinguishes what was observed, what support was used and how uncertain the interpretation is. Age helps select suitable context and language; it must not determine every challenge.

CAST's framework offers useful options for agency and expression [G05]. Recent adaptive-instrument teaching research reinforces the importance of customisation, learner agency and sustained educational support [R14].

# 6. Australian standards and alignment

## Australia has music curriculum expectations

Australian Curriculum Version 9.0 includes Music within The Arts. It uses Foundation and two-year bands through Year 10. Its four connected strands cover exploring/responding, developing practice/skills, creating/making and presenting/performing [C01].

ACT public schools currently teach Version 9.0 from kindergarten to Year 10 [C08]. NSW uses its own syllabuses: the new primary Creative Arts syllabus begins implementation in 2027, while the new Music 7-10 syllabus begins in 2026 [C09-C10]. Years 11-12 require jurisdiction-specific course mapping [C11].

## Checked primary code map

Codes below identify content descriptions, not complete achievement standards or app certification. Summaries are paraphrases. Sources: Foundation [C03], Years 1-2 [C04, C12], Years 3-4 [C02], Years 5-6 [C05].

| Band | Context and response | Skills | Making | Performance |
|---|---|---|---|---|
| Foundation | AC9AMUFE01 | AC9AMUFD01 | AC9AMUFC01 | AC9AMUFP01 |
| Years 1-2 | AC9AMU2E01; AC9AMU2E02 | AC9AMU2D01 | AC9AMU2C01 | AC9AMU2P01 |
| Years 3-4 | AC9AMU4E01; AC9AMU4E02 | AC9AMU4D01 | AC9AMU4C01 | AC9AMU4P01 |
| Years 5-6 | AC9AMU6E01; AC9AMU6E02 | AC9AMU6D01 | AC9AMU6C01 | AC9AMU6P01 |

## What the app could contribute

- Context: selected listening material and discussion of musical purposes.
- Skills: listening activities and deliberate control of musical elements.
- Making: student-authored compositions and documented revisions.
- Performance: rehearsed work shared with a teacher, peers or another appropriate audience.

The E02 descriptions concern First Nations music and contexts; they need substantive, culturally appropriate teaching. A token song or an AI-written story does not establish coverage.

The exact codes were cross-checked across official documents because some support sheets contain apparent typographical inconsistencies. Pin source version and jurisdiction in the content system and obtain educator review.

## A responsible claim

Use wording such as: **Designed to support selected Australian Curriculum Version 9.0 Music content descriptions across Foundation-Year 6.** Publish the actual coverage map once lessons are reviewed.

Every alignment entry should connect a description to a learning intention, activity, observable evidence, teacher rubric and known gaps. A listening-and-keyboard app by itself cannot establish complete music achievement, including the full vocal, social, cultural and performance dimensions.

# 7. Feedback that teaches

## An illustrative learning episode

The student wants a pause in their musical phrase while maintaining the pulse. A return after the rest repeatedly falls away from the intended beat.

Carmen should:

1. Check that the task, recorded input and timing reference are reliable.
2. Replay the student's actual phrase alongside an optional model.
3. Invite the student to identify the difference.
4. Offer a brief listening comparison or a simpler input option if the barrier is unclear.
5. Let the student choose a strategy: slower tempo, beat-only practice or a shorter phrase.
6. Give one specific observation after a phrase, then invite a retry.
7. Use a fresh example without teaching hints to check understanding.

This is a proposed interaction, not a reported trial result. Off-screen singing or clapping is not automatically observable; teacher observations must be recorded separately.

## Three types of feedback

**Task feedback:** identify a supported observation, such as the return entering before the reference beat.

**Strategy feedback:** suggest a purposeful action, such as keeping the pulse through the silence, replaying the entry or practising the transition.

**Self-regulation feedback:** ask the student to select a focus, compare attempts or explain which strategy helped.

Start with one priority. Provide additional detail when requested. Avoid interruption on every note and vague praise that implies improvement without evidence.

## Differentiate intent and task constraints

In an exact imitation task, a changed rhythm is a mismatch with the requested model. In an expressive or composing task, a timing difference may be deliberate. The system should use the declared task, invite the learner's account and defer artistic judgement when evidence is insufficient.

Algorithmic analysis cannot reliably infer expressive intent from timing alone. Nor does a generative model gain that ability merely by being told to act like a teacher.

## Preserve independence

Use optional cues, demonstrations and explanation before answer delivery. Require a learner attempt when appropriate, but permit exploration without a test. Record supported and independently completed work separately.

R03-R07 motivate these choices within their stated limits. The exact combination still needs direct evaluation in Carmen. The aim is for the learner to become more capable of listening and deciding, rather than increasingly dependent on feedback.

# 8. Authentic instruments and credible AI

## Separate musical sound from AI inference

Use sample-based piano and percussion with believable attacks, releases, dynamics and articulations. The AI proposes musical actions or response plans; it does not need to synthesise acoustic waveforms. A typical touchscreen does not reliably measure piano-style touch velocity, so provide intentional dynamic controls.

| Layer | Responsibility | Boundary |
|---|---|---|
| Instrument engine | Immediate sampled sound and accurate note release. | Never wait for inference. |
| Event analysis | Facts about the actual virtual-instrument attempt. | No invented notes, physical-technique or emotional claims. |
| Local learned planner | Propose a bounded response related to the phrase and goal. | Actual model inference and provenance are required. |
| Compiler/verifier | Build and inspect the permitted symbolic transformation. | Musical validity does not prove teaching effectiveness. |
| Reviewed feedback | Explain one supported point and offer choices. | Do not invent curriculum or diagnose causes. |
| Teacher | Review evidence, sequence and interpretation. | AI is not the final assessor. |

## A feasible initial route

Begin with brief replies between phrases. Use a compact local model that proposes reviewed operations, or evaluate a suitable small symbolic music model. Learned operation selection must be called planning; do not present it as unrestricted note generation.

Browser note-model precedents exist, but current dependencies, checkpoints and rights need a real feasibility spike [U16-U17]. Newer local audio generation also exists; its documented hardware route is not a universal school-browser solution [U18-U20].

## Offline and failure behaviour

Child activity stays on the device. No external AI API is authorised for the MVP. Bundle verified essential resources before claiming offline readiness. When AI is unavailable or its plan fails checks, use accurately labelled reviewed support. The learner can continue playing and saving.

## Optional movement

A local camera model may control one expressive parameter with equivalent touch/keyboard access. This is not a unique feature. It must not harm audio timing, require sustained uncomfortable movement or become precision rhythm assessment. Measure the actual device path; specification features are not benchmarks [U21; U24].

# 9A. What already exists

## The nearest documented overlaps

The full comparison file has 29 entries. This selection explains the changes to Carmen's claim.

| Comparator | Documented overlap | Consequence |
|---|---|---|
| Moises / MusicFirst [U01-U02] | Learner-conditioned generated parts and an education offer. | An AI bandmate is not a sufficient USP. |
| Soundverse [U07] | Teaching formats with short musical examples. | Generated educational examples alone are insufficient. |
| EarMaster [U08] | Adaptive questions and musical call-and-response. | Adaptive musical exercises are established. |
| Musiah [U06] | Prioritised AI piano coaching. | Meaningful feedback is a competitive requirement. |
| Maru [U09] | Browser camera control. | Gestures are not new; availability is private early access. |
| AUMI / Clarion [U10; U40] | Adaptable musical participation and teaching support. | Accessibility needs measured implementation value. |
| MIROR [U11] | Children learning through imitation, variation and movement. | The pedagogical interaction has research precedents. |
| Metronaut [U04] | Performance-following accompaniment. | Tempo following is established; platform limits matter. |
| MusiQuest / Charanga [U30; U39] | Creative learning and teacher workflows. | A coherent school experience has close competitors. |
| Magenta RealTime 2 [U18-U20] | Local interactive music generation. | Local AI generation is not a market-first claim. |

## Availability is not uniform

Violy's cited announcement concerns a particular market [U05]. EyeLa is a research installation; ANU's AR work is a research opportunity [U14-U15]. These are relevant precedents without being interchangeable school products.

Product claims were not authenticated or measured hands-on. No missing feature is labelled absent merely because a public page did not mention it. Marketing exclusivity and learning-speed claims were not adopted.

# 9B. A conditional selling proposition

## The claim Carmen may be able to earn

Carmen aims to help a mixed-ability primary group learn through student-owned musical experiments: a local AI proposes a goal-related reply, the system verifies what it changed, and the learner demonstrates a related idea independently.

No reviewed material establishes that workflow as globally exclusive. The commercial question is whether it is better for a defined customer than a reasonable existing workflow.

## What to demonstrate

- A response materially depends on the student's actual phrase and chosen objective.
- Different evidence leads to different, justified support.
- A response breaking the stated musical constraint is rejected.
- Students can redirect the partner while keeping their original work.
- An independent check supports a limited, interpretable learning claim.
- Teachers can guide the activity without excessive preparation.
- The same architecture supports distinct primary pathways and an older extension.

## What to compare

For foundations, use a strong Musicca/EarMaster/teacher activity. For creative work, examine a suitably licensed Moises/MusicFirst or other realistic creation workflow. Compare content, time, access adjustments and human help fairly.

Separate the contribution of adjustable controls, reviewed teaching rules and model planning. If rules produce the same result more reliably, improve the model or narrow the AI claim. If an existing tool makes the workflow easier or cheaper, revise the proposition.

## Potential longer-term advantages

A reviewed task/constraint library, good teaching decisions, reliable integration, inclusive usability and trusted implementation can become valuable assets. A mascot, generic AI prompt or stock model is easy to imitate. Child data is not the moat.

Before a funding claim, establish a useful outcome, practical delivery cost, a budget owner and a repeat purchase reason. Grants and hackathon awards cannot establish recurring customer demand.

# 10. The working MVP across primary school

## Preserve the full suite scope

Use a common event, objective and response contract across Foundation-Year 6. Keep the ten original draft foundations and add distinct starter missions for every band.

| Pathway | Proposed working example | Evidence |
|---|---|---|
| Foundation | Explore and contrast sounds in an original sound story. | Intentional choices and suitable adult-supported response. |
| Years 1-2 | Create a rhythmic conversation and investigate a rest. | Listening and live timing reported separately. |
| Years 3-4 | Develop a melodic reply while preserving a time pattern. | A controlled musical decision and fresh application. |
| Years 5-6 | Organise motifs into a chosen form and explain revisions. | Documented structural choices and independence. |
| Older extension | Critique an arrangement and defend an alternative. | Analytical depth without compulsory higher motor demands. |

These mission cards are drafts. Complete sequential coverage and educator review remain a content programme. Do not populate unimplemented year levels with screens that imply finished courses.

## The convincing demonstration

Show an original learner phrase, a real model-proposed response plan, the verified musical result, student direction and a fresh independent task. Then show the teacher the original contribution, assistance and evidence.

A rules-only reply can demonstrate the interface, but it cannot satisfy the claimed model path. An optional gesture recogniser does not substitute for meaningful musical AI.

## Build order

First verify the local-model and device route. Build direct sampled sound and a complete interaction. Then develop the four meaningful primary pathways, teacher evidence, original creation, offline behaviour and the older extension. Add the optional gesture experiment only if it helps participation and preserves audio responsiveness.

No weights, samples, app, child outcomes or educator approvals are supplied by this research pack. The implementation agent must produce and inspect actual working evidence.

# 11. Teacher Mode and inclusive deployment

## Give the teacher decisions they can act on

A useful teacher view answers:

- What was this student trying to learn?
- What did the system actually observe?
- What support was active?
- What is still uncertain?
- What short activity or human intervention might help next?

A single percentage cannot answer these questions. The teacher should be able to replay an attempt, inspect the task target, change a recommendation and record their own observation.

## A workable primary-school workflow

Choose a project and learning intentions. Assign an appropriate starting version to a group. Offer separate controls for musical material, representation, interaction and teaching cues. Review a small number of useful patterns across the class, then lead a group activity or rehearsal.

Create a common musical outcome with different contributions: beat, rhythmic response, melody, texture or form. Shared composition can initially use recorded parts or a teacher-led common pulse. Reliable live synchronisation across many browsers is a separate engineering challenge.

## Include teaching support

The product needs short demonstrations, explanations of the musical goal, alternative ways to teach it and a suggested group activity. Recent adaptive-instrument design research identifies pedagogical customisation, learner agency and community integration as important alongside the instrument [R14]. Open Orchestras is a practical precedent for packaging instrument access with repertoire and training [M11].

## Accessibility and equity

Provide keyboard navigation, visible focus, accessible names, alternatives to dragging and adequate targets. Use WCAG 2.2 AA as a testable technical target [G04], alongside direct testing of the instruments and musical tasks.

Allow a reduced-motion view, uncluttered controls, adjustable visual density and captions or text alternatives to instruction. Avoid assuming a particular impairment from input behaviour. Different hearing and motor needs may require different learning objectives or evidence, agreed with the teacher and learner.

Device access, bandwidth, audio output and teacher capacity remain real constraints. Offline preparation and shared-device support help only when a workable device and adult setup are available.

## Child and school controls

Use minimal profiles, role-restricted access, explicit retention/deletion choices, appropriate vendor terms and no advertising or behavioural targeting. Keep student creations under clear ownership and permission arrangements. Teacher override must affect the actual learner experience, not merely add a note to a report.

# 12. Why schools, government or universities might invest

## Treat each institution as a different customer

| Stakeholder | A credible reason to consider Carmen | Evidence they would need |
|---|---|---|
| Generalist teacher / school | A teachable primary sequence and practical differentiation without a personal instrument for each child. | Setup time, classroom usefulness, musical learning and appropriate cost. |
| School leader / curriculum coordinator | A consistent progression with transparent coverage and evidence. | Educator review, usable assessments, training needs and staff adoption. |
| Education authority / government programme | Better musical participation or learning for a defined underserved group at a sustainable delivery cost. | Comparative outcomes, equitable access, privacy/security, accessibility and a credible implementation plan. |
| University education or HCI researchers | A researchable environment for studying musical understanding, adaptive interaction and learner agency. | Research protocol, valid measures, ethical access and transparent limits. |
| University outreach programme | A school/community programme combining creative learning with measurable participation and human pathways. | Partner demand, repeat delivery and an affordable support model. |

These are adoption hypotheses, not evidence that any university or government wants to purchase Carmen. A research partner, grant funder and recurring customer play different roles.

## The need is plausible, but test the exact job

The NSW primary survey identifies resource, preparation and professional-learning needs [N01]. Its associated journal abstract reports 706 teachers from 95 schools [N02]. These are the same underlying research programme, and NSW findings should not be converted into a national market estimate.

Ask potential buyers to show the lesson they currently use, the adjustments they make, the budget that could pay and the decision process required. Test Carmen against the full existing workflow, including good free resources.

## Adoption requirements

The Australian generative-AI framework supports teacher expertise, critical thinking, fairness and human accountability [G01]. ST4S publishes a vendor assessment guide, with a current 2026.1 version [G02]. Its responsible-AI work provides additional education-focused controls [G03].

These are relevant adoption frameworks, not proof that a product improves learning or blanket national procurement permission. Product eligibility and jurisdiction requirements must be checked.

OAIC materials retrieved for this review describe development of the Children's Online Privacy Code and registration required by 10 December 2026 [G06]. Check final status, applicability and commencement before deployment. Keep the product's data design conservative in the meantime.

# 13. Evaluate the learning and the technology

## Separate the effects

Compare a fixed instrument with reviewed feedback, an adjustable instrument with reviewed response rules, and the same environment with learned response planning. This avoids crediting AI for an improvement caused by clearer teaching or better controls.

Compare gestures with equivalent accessible controls separately. Do not remove access adjustments to create an independent check.

## Learning and implementation measures

| Question | Useful evidence |
|---|---|
| Does understanding carry over? | A fresh task and delayed application with teaching hints removed. |
| Does the model help? | Appropriate response selection versus a reviewed rules baseline. |
| Does the system respect the music? | Invalid outputs rejected, uncertainty shown and original work preserved. |
| Does the learner direct it? | Meaningful decisions, revisions and ability to reject suggestions. |
| Can different learners participate? | Actual completion/support needs across primary bands and relevant access needs. |
| Can a teacher sustain it? | Preparation, intervention and review time against the previous workflow. |
| Can the service sustain it? | Model/device load, content/support cost and repeat budgeted use. |

## Research stages

Begin with educator review and adult technical/usability checks. Then conduct appropriate formative and feasibility work across the primary bands, with older learners as a distinct group. Later comparative trials need a justified sample, pre-specified outcomes and analysis appropriate to class/school grouping.

A few successful sessions do not establish durable learning or national suitability. Synthetic fixtures and engineering benchmarks are not student research.

## Investment gate

Require a demonstrated user benefit and a credible purchase path. If the model adds no useful value, revise its role and claims. Report failures and subgroup barriers as part of the evidence, rather than turning engagement into a learning score.

# 14. Sustainability and investment

## The buyer has to value the result

Schools may value a teachable, inclusive sequence and manageable differentiation. A university may value a researchable interaction or an outreach programme. Government programmes may value equitable participation at an acceptable cost. These are hypotheses until a budget owner and recurring need are identified.

## Costs beyond the model

Content authorship, educator review, cultural permissions, accessibility, device support, security, teacher training and school onboarding continue after a prototype works. Local inference avoids a per-turn vendor charge in the default route but does not make delivery free.

| Risk | Practical response |
|---|---|
| A model needs expensive/new hardware. | Test ordinary target devices before integrating it. |
| Inference interrupts playing. | Keep audio independent and use bounded replies between phrases. |
| A lesson library becomes costly to maintain. | Use reviewed task families, versioned sources and an editorial process. |
| Gestures add fatigue or exclusion. | Keep them optional with equivalent controls and measured benefit. |
| AI changes the wrong musical property. | Validate explicit symbolic invariants and abstain when necessary. |
| An existing product is sufficient. | Compare fairly and revise the proposed advantage. |

## A commercial route to test

A budgeted school or outreach pilot can test implementation and willingness to pay. A later annual licence could fund reviewed content, support and maintenance. No price, market size, margin or return on investment is established here.

## Scope and release honesty

Whole-primary architecture is required from the outset. A complete curriculum catalogue, safe school deployment and comparative evidence take further work. Do not confuse a narrow working prototype with a completed suite, or an unimplemented vision with a release.

The technical proposition should survive a fair comparison with existing products. Its investment case must rest on the learner/teacher outcome and sustainable delivery, not a claim that AI or AR is inherently innovative.

# 15. Model and device decisions

## Current evidence changes what is feasible

Magenta's June 2026 RealTime 2 release demonstrates local live music synthesis. Its published small/base variants use different Apple-Silicon hardware routes [U18-U20]. That is useful current evidence, but not support for promising browser deployment on ordinary school devices.

| Route | Initial decision | Evidence required |
|---|---|---|
| Compact local learned planner | Smallest initial AI target. | Real inference, rights, phrase/goal conditioning and baseline comparison. |
| Local symbolic music model | Candidate feasibility spike. | Exact checkpoint, runtime, dependencies and target-device measurements. |
| Large live audio model | Research comparator. | A separate hardware/product case; no universal browser assumption. |
| Browser hand recognition | Optional experiment. | Worker scheduling, usable mapping, no image transfer and equivalent controls. |
| Immersive AR | Later research. | A specific learning/access benefit and workable hardware cost. |

A runtime supporting WASM or WebGPU does not establish that a particular model works acceptably [U22]. A repository code licence does not automatically establish model-weight or training-data rights.

## Local requirement

No cloud AI or uploaded child session is authorised by this update. If the model route fails, preserve the guided core and label the AI path experimental or blocked. Do not use a hidden remote service or a mock response to make a demonstration appear complete.

## Technical acceptance

The real implementation must include a model card, exact versions/hashes, held-out and rules-baseline comparisons, cold/warm timings, memory/download measurements, fallback rate and offline/network observations. The research pack contains no such executed results.

# 16. Budgets and operational gates

## Proposed targets, not measurements

Initial targets include responsive direct sound, warm replies within about two seconds and a bounded timeout. A proposed p95 end-to-end sound target of 50 ms is a design goal; a fast callback does not prove acoustic latency. Record device, browser, output route, cold/warm state and measurement method.

Keep first-use downloads compact and offer richer optional packs. Measure actual model memory, battery/thermal behaviour and audio stability. Do not infer sustainability from parameter count.

## Model-call economics

The default local route has no per-turn vendor API bill. For a possible later cloud route, model the complete cost rather than quoting an unlimited consumer plan.

Example assumptions only: 200 learners x 20 sessions x 12 turns = 48,000 turns per period. At assumed costs of A$0.001, A$0.005 or A$0.02 per turn, variable cost is A$48, A$240 or A$960. These are sensitivity calculations, not vendor prices or a cloud-processing decision. Include retries, hosting, support, content amortisation and onboarding.

## Practical sustainability

Do not run vision or generation continuously when a short musical interaction is sufficient. Pause and release resources when inactive. Local processing is not inherently lower carbon, and no environmental advantage is quantified.

Prioritise musical and educational reliability over a large number of unsupported instruments or agents. Maintain component boundaries so a model, runtime or content pack can be replaced without losing learner work.

## Continue only with evidence

A model integration, gesture feature or larger curriculum release should have an explicit benefit and a support plan. Use real primary educators and relevant learners to identify where the proposed system fails. The market comparison, classroom implementation and long-term learning questions remain open.

# 17. What Claude Code receives

## Implementation documents

The updated archive contains concise project instructions, the complete master context, the novelty audit, a 29-entry comparison matrix and the research search log. Technical files describe modes, response planning, musical verification and device/cost gates.

## Structured draft data

- A primary-suite plan covers all four bands and an older extension.
- The ten original foundation lesson drafts are retained.
- A response-plan schema specifies bounded model output.
- Illustrative examples contain valid and invalid symbolic transformations.
- A model intake record is deliberately unapproved.
- A claim register distinguishes positioning, rejected novelty claims and gated capabilities.

These files are not a trained model, shipped curriculum or measured student result.

## Instructions that prevent common implementation errors

Use authentic samples for audio. Keep original input and assisted output separate. Do not let a model alter a score or assert a diagnosis. Do not treat a generated note as a note the student chose. Reject invalid musical replies before playing them. Keep learner authority and teacher judgement. Preserve the local-data requirement and optional camera boundary.

## First meaningful build

Inspect the repository, test the model/runtime route and build one complete interaction. Then implement the distinct primary pathways and older extension using the common engine. Expand the reviewed content rather than renaming one lesson for every age.

The pack's validators inspect data consistency and fixture relations. They do not certify an app, model, school deployment, accessibility compliance or market uniqueness.

# 18. Research scope and confidence

## What was reviewed

This update used 31 targeted queries and direct primary-source retrieval. The matrix distinguishes products, school programmes, private early access, research installations and model tooling. Official documentation provides stronger feature attribution than unsourced comparison articles. It still does not establish that a feature works as described.

## What was not done

No authenticated paid-product trial, complete patent search, model installation/benchmark, real-child test, educator approval or procurement negotiation occurred. The findings therefore support overlap and feasibility decisions, not an exclusive market claim.

## Particular limitations

The cited Violy announcement is market-specific. Maru identifies private early access. MusiQuest's direct page was unreadable; official indexed material and its curriculum PDF were available. Yousician's direct page returned 403 while official indexed material was available. A redirected Quaver Health/PE page was excluded; the music comparison uses a relevant official review.

Credit documentation can be inconsistent. No current vendor price or credit entitlement is relied upon for Carmen's unit economics.

## Source handling

The archive's merged source catalogue preserves original IDs and identifies URL aliases. Multiple pages about one product or papers from one research programme are not independent evidence. Review depth and limits are recorded; a source used for research is not approved as a reusable product asset.

## The honest conclusion

None of the proposed components alone is established as a unique selling point. The candidate value lies in an implemented and evaluated connection between accessible music-making, bounded local AI response planning, musical verification, student agency and useful teacher evidence. The next step is a working comparison, not a stronger adjective.

# Sources A: learning and pedagogy

**R01 | McPherson (2005).** From child to musician: skill development during the beginning stages of learning an instrument. Longitudinal study; 157 beginning instrumentalists over three years. Publisher abstract reviewed; not a causal test of Carmen.

**R02 | Wan, Crawford & Jenkins (2022).** Facilitating self-regulation of instrumental practice with digital technology. Semi-systematic review, Australian Journal of Music Education 54(2), 83-98. Full text reviewed; older literature and heterogeneous designs.

**R03 | Wan, Crawford & Jenkins (2023; online 2022).** Digital listening tools to facilitate children's self-regulation of instrumental music practice. Four piano learners aged 8-10 and one teacher, 22 weeks. Author-institution abstract reviewed; qualitative cases, no control group.

**R04 | Hendriks et al. (2023).** Promoting creative autonomy support in school music education. Intervention involving 18 teachers and 105 lessons in six primary schools. Full text and corrigendum reviewed; evaluates teacher interaction, not an AI app.

**R05 | Blanco, Tassani & Ramirez (2021).** Real-Time Sound and Motion Feedback for Violin Bow Technique Learning. Randomised feedback study with 50 analysed novices, plus expert comparison. Full text reviewed; short, specialised physical-instrument task.

**R06 | Peng et al. (2026).** Empowerment or dependency? Intelligent assessment, generative AI and cognitive agency in music education. Systematic review of 21 empirical studies. Full text reviewed; mostly Chinese/higher-education settings, scarce longitudinal and younger-child evidence.

**R07 | Bastani et al. (2025).** Generative AI without guardrails can harm learning: Evidence from high school mathematics. Field experiment with nearly 1,000 high-school students. Full text and affiliation correction reviewed; mathematics, not music.

**R08 | Sala & Gobet (2020).** Cognitive and academic benefits of music training with children: A multilevel meta-analysis. Meta-analysis of non-music outcomes. Full text reviewed; cautions against claiming broad cognitive/academic transfer from music participation.

**R09 | AERO (current guide).** Explicit instruction: Practice guide. Australian education evidence guidance. General teaching research; not validation of a touchscreen music lesson sequence.

**R10 | AERO (current guide).** Scaffold practice. Guidance on supports and gradual independence. Apply to teaching hints while preserving access adjustments.

**R11 | AERO (updated 2026).** Vary practice. Guidance on spacing, variation and retention. General evidence; specific musical schedules remain a design choice.

# Sources B: adaptive instruments and curriculum

**R12 | Pardue, McPherson & Overholt (2018).** Improving the instrumental learning experience through complexity management. Research position paper; author-institution abstract reviewed. Strong precedent for adjustable instrument complexity, not proof of Carmen efficacy.

**R13 | Grond, Shikako-Thomas & Lewis (2020).** Adaptive Musical Instruments: Past, Present, and Future Research Directions. Review of instruments for disabled children; journal abstract reviewed. Warns that improvisatory and social value cannot be reduced to convenient output scores.

**R14 | Bremmer et al. (2026).** Boundless Grooves: Designing One-to-One Lessons for Adaptive Digital Musical Instruments. Educational design research, six initial teachers and seven testing teachers. Publisher abstract reviewed; supports agency, customisation and teaching/community support.

**P01 | Kodaly Queensland.** The Kodaly approach. Professional association account of singing, sequencing and sound before symbols. A pedagogical tradition, not comparative efficacy evidence.

**P02 | West Australian Orff-Schulwerk Association.** About WAOSA & Orff. Professional association account of imitation, exploration, improvisation and composition. Does not establish superiority over other approaches.

**P03 | Gordon Institute for Music Learning.** Audiation. Source description of musical thinking and tonal/rhythmic understanding. Audiation is not directly measurable from touchscreen accuracy alone.

**P04 | Bremmer & Nijs (2020).** The Role of the Body in Instrumental and Vocal Music Pedagogy. Theoretical synthesis of embodied teaching. Supports attention to movement and interaction; not an app-effectiveness trial.

**C01 | ACARA, Version 9.0.** The Arts: learning-area information, Music. Official national curriculum structure and rationale. Jurisdictions and schools determine implementation.

**C02 | QCAA / ACARA.** Years 3-4 Music: achievement standard aligned to content descriptions. Official Version 9.0 mapping; pilot codes checked against the PDF. Descriptions in this report are paraphrases.

**C03 | QCAA / ACARA.** Foundation Music: achievement standard aligned to content descriptions. Official Foundation progression reference; play, ideas and sharing. Not a preschool or complete teaching programme.

**C04 | QCAA / ACARA.** Years 1-2 Music: achievement standard aligned to content descriptions. Official progression reference. Apparent code inconsistencies in this sheet require checking against ACARA before reusing individual codes.

**C05 | QCAA / ACARA.** Years 5-6 Music: achievement standard aligned to content descriptions. Official reference for expressive choices, composition, documentation and performance. Used for whole-primary planning; not a complete reviewed unit.

# Sources C: progression and school adoption

**C06 | QCAA / ACARA.** Years 7-8 Music: achievement standard aligned to content descriptions. Official secondary reference for analysis, reflection, composition and performance. A separate content and assessment phase.

**C07 | QCAA / ACARA.** Years 9-10 Music: achievement standard aligned to content descriptions. Official reference for stylistic interpretation, evaluation and composition. Does not establish senior-secondary alignment.

**C08 | ACT Government.** What we teach: curriculum. Current ACT public-school page states Australian Curriculum Version 9.0 is taught from kindergarten to Year 10.

**C09 | NESA.** Creative Arts K-6 Syllabus (2024): overview. Implementation from 2027; preparation in 2026, with early implementation permitted. A NSW product needs its own mapping.

**C10 | NESA.** Music 7-10 Syllabus (2024): overview. Implementation from 2026. Consult sector transition advice; national-code mapping alone is insufficient in NSW.

**C11 | ACT BSSS.** Music A-T-M-V. Senior-secondary course reference. Future Year 11-12 claims need a separate course-level assessment mapping.

**C12 | QCAA / ACARA.** Prep-Year 6 Music: sequence of content descriptions. Official full-primary sequence; confirms Year 1-2 performance code. Contains an apparent Foundation development-code omission; cross-checked with C03.

**N01 | Taylor & Edwards / ACER (2025).** Understanding the delivery of music learning in NSW public schools. Primary teachers' survey report. Resource, preparation and professional-learning needs; repository sample wording conflicts with the journal abstract.

**N02 | Taylor & Edwards (2026; online 2025).** Music Education in NSW Government Primary Schools: Teachers' Experiences, Training and Perceptions. Publisher abstract reports 706 teachers from 95 schools. Same underlying survey as N01, not an independent replication; NSW evidence.

**G01 | Australian Government Department of Education.** Australian Framework for Generative AI in Schools. Official page and downloadable framework reviewed. Review endorsed June 2025; guidance is not product certification or proof of educational efficacy.

**G02 | Education Services Australia / ST4S.** ST4S Vendor Guide. Current listed guide: 2026.1, updated 21 July 2026. Assessment scope and eligibility need checking; a readiness check is not approval.

**G03 | Education Services Australia / ST4S.** The Responsible AI Standard. Education-specific AI controls alongside ST4S. Page describes an invitation-based evaluation pilot after ST4S assessment.

# Sources D: access, audio and alternatives

**G04 | W3C.** Web Content Accessibility Guidelines 2.2. Technical accessibility standard. Conformance requires testing complete flows, and does not itself prove inclusive music pedagogy.

**G05 | CAST (2024).** Universal Design for Learning Guidelines 3.0. Design framework for agency and varied engagement, representation and action. Not an Australian curriculum or automatic accessibility certification.

**G06 | OAIC.** Children's Online Privacy Code. Retrieved materials describe consultation and registration required by 10 December 2026. Verify final status, applicability and commencement before deployment.

**G07 | Creative Australia / Terri Janke and Company.** Protocols for using First Nations cultural and intellectual property in the arts. Authority, consent, attribution and appropriate collaboration. Public availability or copyright permission does not settle cultural permissions.

**T01 | W3C.** Web Audio API 1.1. Official audio timing specification. Reported latency estimates are platform-dependent and do not replace end-to-end measurement.

**M01 | Duolingo.** Duolingo Music Is Here. Official documentation of on-screen keyboard, rhythm, listening and notation learning. Product description, not an independent efficacy audit.

**M02 | Musicca.** Introduction for educators. Official page documents free school tools, virtual instruments, adaptive exercises and progress tracking. No hands-on comparative trial conducted.

**M03 | Charanga Australia.** Yumu for Students. Official student learning and music-making documentation. Existing classroom/home workflow overlaps with Carmen.

**M04 | Charanga Australia.** YuStudio. Official documentation of beginner-friendly music creation and teacher project guides. Does not establish every feature or offline capability.

**M05 | MakeMusic.** Practice. Official interactive practice and feedback documentation; assessment can be disabled. Real-time feedback is already an established product feature.

**M06 | Ableton.** Learning Music. Official browser-based learning and creation; no previous equipment required. Strong comparator for instrument-free musical fundamentals.

**M07 | Google Creative Lab.** Chrome Music Lab: About. Official hands-on experiments and Song Maker; no account required. Do not treat exploratory web music as a novel Carmen invention.

**M08 | Apple Support.** Get started with GarageBand for iPhone. Official touch-instrument, recording and creation documentation. Platform/features do not imply a curricular teaching programme.

**M09 | MusiQuest.** MusiQuest's Lessons and Curriculum. Official account of guided listening, skills, assessment and composition without instrumental prerequisites. Especially close creative-learning comparator.

**M10 | MIT Media Lab.** Hyperscore project overview. Primary project documentation of graphical musical composition. A precedent for expressive creation through alternative representations, not a current market audit.

**M11 | Open Up Music.** Open Orchestras programme and Clarion. Official adjustable instrument, repertoire, training and evaluation offer. Demonstrates why accessible instruments and teaching support already have substantial precedents.

Source notes: all web resources checked 9 November 2026. Curriculum descriptions in this report are paraphrased. Australian Curriculum material is copyright ACARA 2010-present, licensed CC BY 4.0 subject to its terms; QCAA support documents are copyright State of Queensland (QCAA), subject to their stated licences and exclusions. No endorsement is claimed. R04's figure corrigendum and R07's affiliation correction were checked. Recommendations, examples and proposed product behaviours are the report's synthesis, not reported Carmen outcomes.

# Sources update 1: innovation and feasibility

**U01 | MusicFirst Australia.** Moises in MusicFirst classrooms. Generated instrumental accompaniment for classroom use is documented. School suitability and learning outcomes were not independently tested.

**U02 | Moises.** Moises Studio. Editable generated parts and a natural-language studio agent are documented; feature/availability claims are not hands-on verified.

**U03 | Moises Help.** AI Studio credits. Credit-based audio generation illustrates usage costs; the page contains inconsistent allowance examples. No price or entitlement inferred.

**U04 | Metronaut Help.** Interpret: AI Tempo. Tempo following documented on native apps; this mode is explicitly unavailable on Web in the reviewed instructions.

**U05 | Violy.** AI automatic accompaniment announcement. China-market announcement; global availability, release year and current entitlements are not established.

**U06 | Musiah.** AI piano teacher. MIDI-keyboard requirement and prioritised adaptive feedback documented. Speed-of-learning and world-first marketing claims not adopted.

**U07 | Soundverse.** AI Music Teacher. Lessons, quizzes, paths and short generated teaching examples documented. Child suitability, exact output fidelity and outcomes unverified.

**U08 | EarMaster.** EarMaster for all musicians. Adaptive questioning, sing/clap-back and alternative input interfaces documented. Vendor exclusivity language not adopted.

**U09 | Maru.** Play with your camera. Browser gesture control described; site identifies private early access. General product availability and practical performance unverified.

**U10 | AUMI.** About the Adaptive Use Musical Instrument. Movement-based improvisation and accessible music-making precede Carmen. Outcomes cannot be inferred from product description.

**U11 | Addessi et al. (2020).** MIROR platform overview. Child-led imitation/variation and embodied musical interaction are established research precedents, not invented Carmen capabilities.

# Sources update 2: innovation and feasibility

**U12 | IRCAM.** Somax2. AI co-improvisation system; does not establish an ordinary-browser primary-school deployment.

**U13 | Google Magenta (2018).** Piano Genie. Learned eight-button-to-piano mapping already exists; generated pitches are not evidence the learner selected those pitches.

**U14 | Cheng and Pang (2026).** EyeLa. Accessible AR music/rhythm interaction using head/eye movement described; not a generalised effectiveness trial.

**U15 | ANU School of Computing (2026).** Spatial intelligent musical instruments in AR. Research opportunity, not an established school product, partnership or evaluated Carmen deployment.

**U16 | Magenta.** Magenta.js repository. Repository states Apache-2.0 code licensing. Maintenance, dependencies, model weights and training rights need separate verification.

**U17 | Magenta.** Magenta Music README. Note-based browser models include MusicRNN/MusicVAE. Legacy examples demonstrate precedent, not current target-device readiness.

**U18 | Magenta Team (4 June 2026).** Magenta RealTime 2. Local interactive audio generation exists; published Apple-Silicon route does not demonstrate browser/tablet/Chromebook compatibility.

**U19 | Magenta.** RealTime 2 model card. Apache-2.0 code, CC BY 4.0 weights and additional terms; rights and responsibilities remain distinct from app suitability.

**U20 | Magenta.** RealTime model hardware requirements. 230M and 2.4B variants have different Apple-Silicon requirements. These are publisher reports, not Carmen measurements.

**U21 | Google AI Edge.** Gesture recognizer for Web. Browser gesture inference available; synchronous calls can block the UI. Worker scheduling and model/data licensing need checking.

**U22 | ONNX Runtime.** ONNX Runtime Web. WASM and GPU execution providers differ by browser and model. A runtime cannot guarantee that a selected model runs acceptably.

# Sources update 3: innovation and feasibility

**U23 | MDN.** WebXR Device API. Immersive web support and hardware constraints require feature detection; do not make XR essential for the access proposition.

**U24 | W3C.** Web Audio API. Scheduling and latency estimates require actual device validation; audio remains independent of inference.

**U25 | Versilian Studios.** Versilian Community Sample Library. CC0 candidate collection; exact files, quality, revision and hashes still need import review. Not every instrument has the requested layers.

**U26 | Texas Education Agency / IMRA.** QuaverMusic Texas Grade 1 review. Documents differentiated teaching and varied assessment in an existing primary suite. It is not an Australian alignment or efficacy trial.

**U27 | BandLab Help.** SongStarter AI. Editable AI song ideas already exist. Do not infer suitability for primary children or reuse rights beyond the actual terms.

**U28 | BandLab Help.** Underage Individual policy. Public-account minimum-age restrictions vary by jurisdiction. This public product is not automatically a primary-school deployment option.

**U29 | Soundtrap.** Soundtrap for Education. School assignments, collaboration and music production documented. Availability does not establish every proposed Carmen feature.

**U30 | MusiQuest.** How it works. Composition, curriculum and score/MIDI export described. Related official curriculum PDF also reviewed; full product not tested.

**U31 | Duolingo.** Music course. Instrument-free keyboard, listening and notation learning documented; older announcement is not a complete 2026 feature inventory.

**U32 | Musicca.** Introduction for educators. Free school tools, virtual instruments, adaptive exercises and progress tracking documented.

**U33 | Yousician.** Company overview. Audio recognition and real-time musical feedback documented. No learning-speed or adoption figures adopted.

# Sources update 4: innovation and feasibility

**U34 | Simply.** Simply Piano. Tailored lessons and performance feedback documented. General lesson claims do not establish every XR feature.

**U35 | MakeMusic.** Practice. Performance feedback, accompaniment, teacher sharing and assessment-off options documented. Exclusivity language not adopted.

**U36 | Apple.** GarageBand for iOS. Touch instruments and music creation documented; not a complete mapped Australian music course.

**U37 | Ableton.** Learning Music. Browser-based fundamentals and composition without previous equipment documented.

**U38 | Google Creative Lab.** Chrome Music Lab. Account-free musical experiments and Song Maker documented.

**U39 | Charanga Australia.** YuStudio. School creation projects, teacher guides, assignments and review of student work documented.

**U40 | Open Up Music.** Open Orchestras / Clarion. Customisable instrument plus repertoire, training and evaluation documented; accessibility alone is not new.

**U41 | MIT Media Lab.** Hyperscore. Graphical composition precedent; not a current paid-product audit.